

Classification of Chemical Reactions:

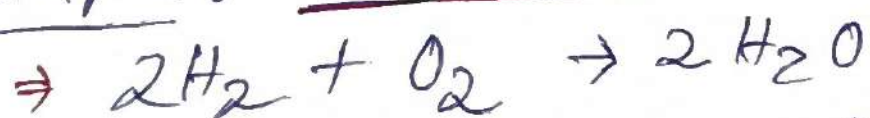
Based on the number and types of phases involved the reactions are:

- a) Homogeneous Reaction
- b) Heterogeneous Reaction.

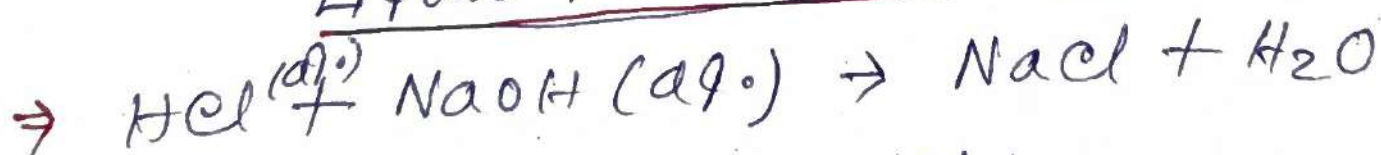
a) Homogeneous Reaction: It is a chemical reaction where all reactants and products are in the same phase (solid/liquid/gas). There are no phase boundaries between different components involved in the reaction.

- * It takes place in one phase alone
- * No phase boundaries among different components.

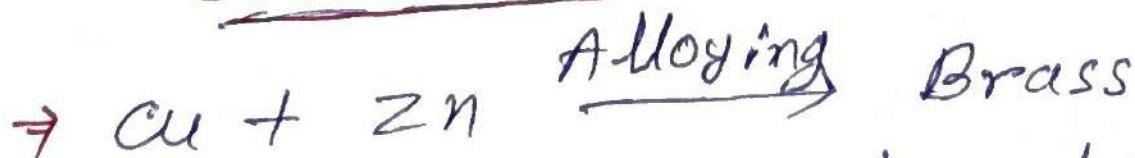
Example: Most gas phase reactions:



Liquid phase reactions:



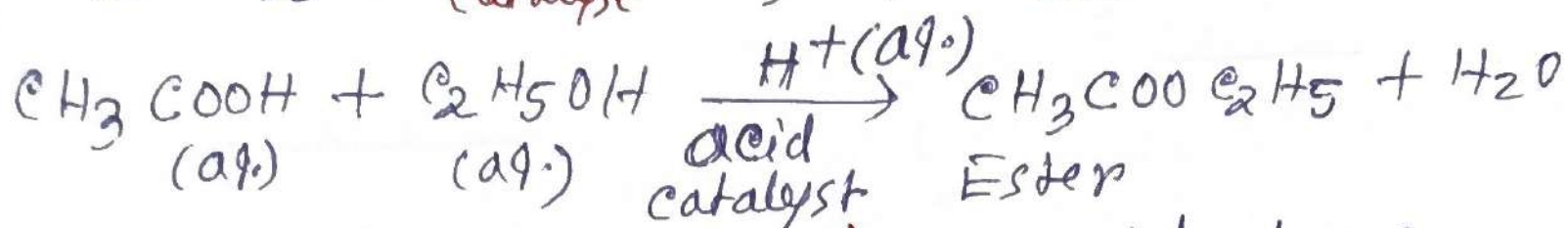
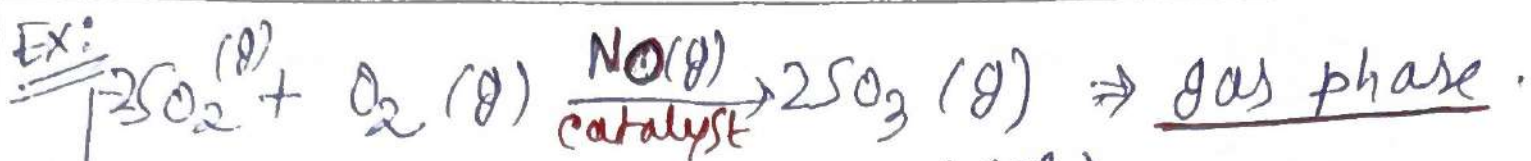
Solid phase reactions:



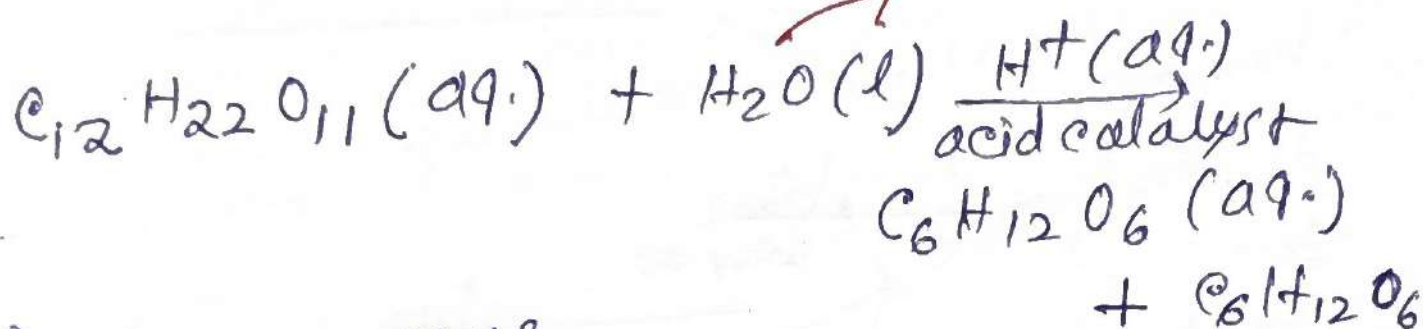
* Homogeneous reaction can be catalytic and noncatalytic.

$\Rightarrow$ If a homogenous reaction is carried out in presence of a catalyst it is called the homogeneous catalytic reaction.

Example: Most liquid phase reaction.



Liquid phase



Homogeneous catalytic reaction.

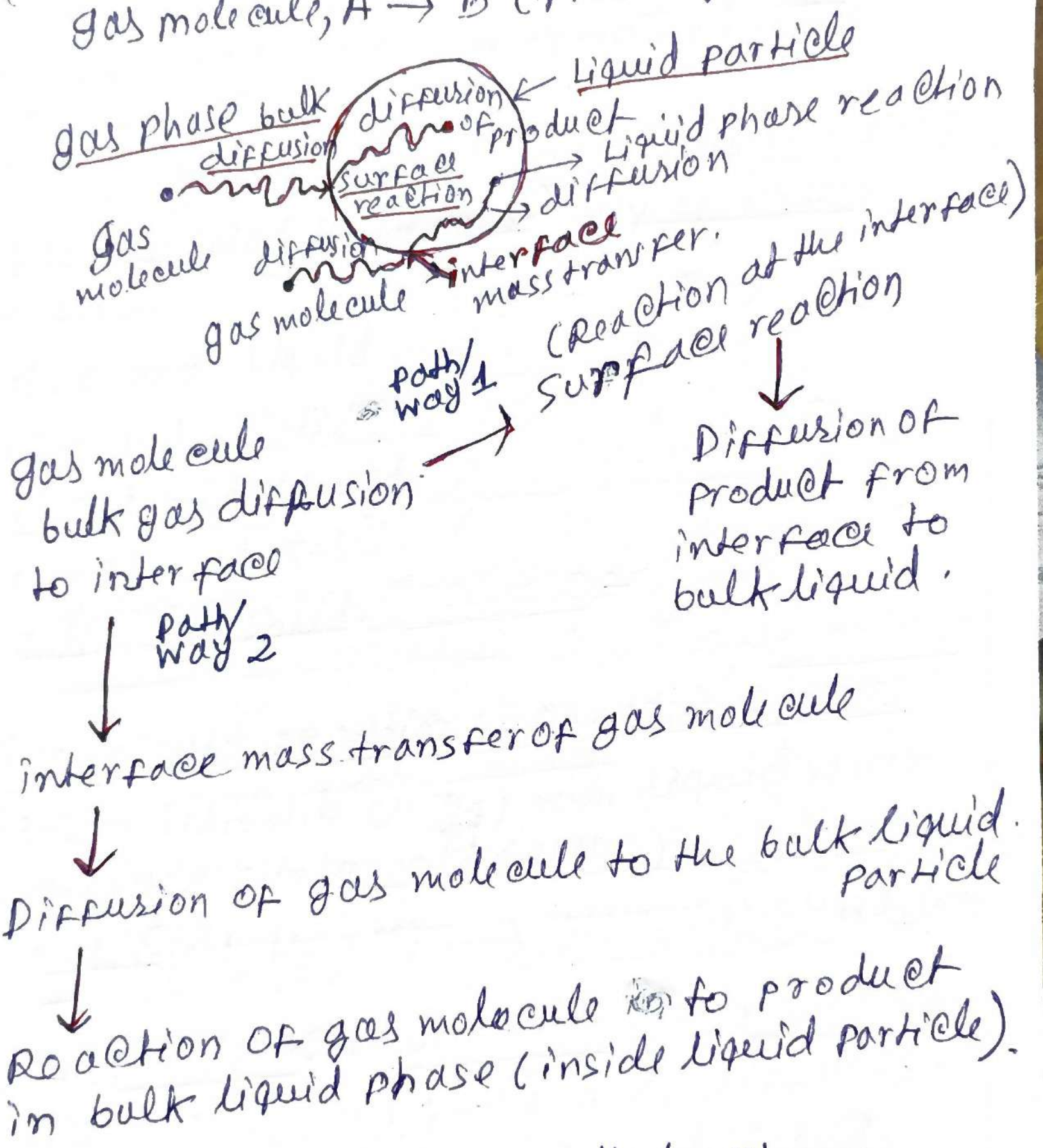
b) Heterogeneous Reaction :

It is a chemical reaction where the reactants exist in two or more different phases (solid, liquid, gas).

- * Reactions involving more than one phase.
- * The reaction takes place either in one phase, two phases or more phases.
- * It may take place at an interface, i.e. at the boundary between two phases.
- * The reactants and products are distributed among the phases or all are contained within a single phase.

Mechanism

Chemical reaction
gas molecule, $A \rightarrow B$ (product)



- It involves two path/way
- ⇒ way 1. ⇒ Through surface reaction
 - ⇒ way 2 ⇒ Through bulk-phase reaction or reaction in a particular phase.

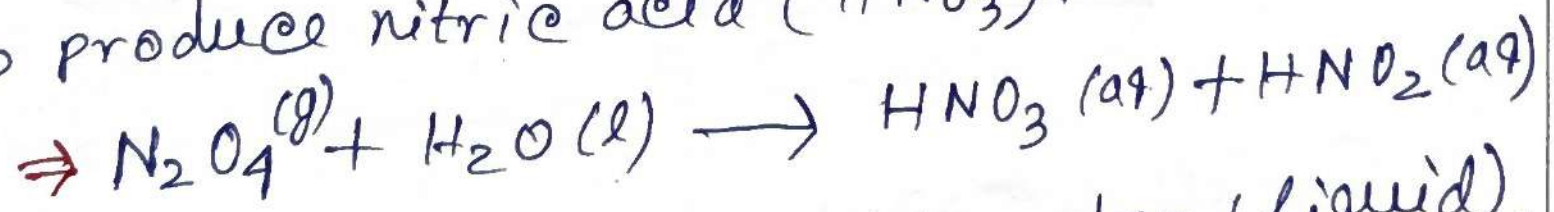
Classification of heterogeneous reactions

It is also classified as catalytic and non-catalytic reactions.

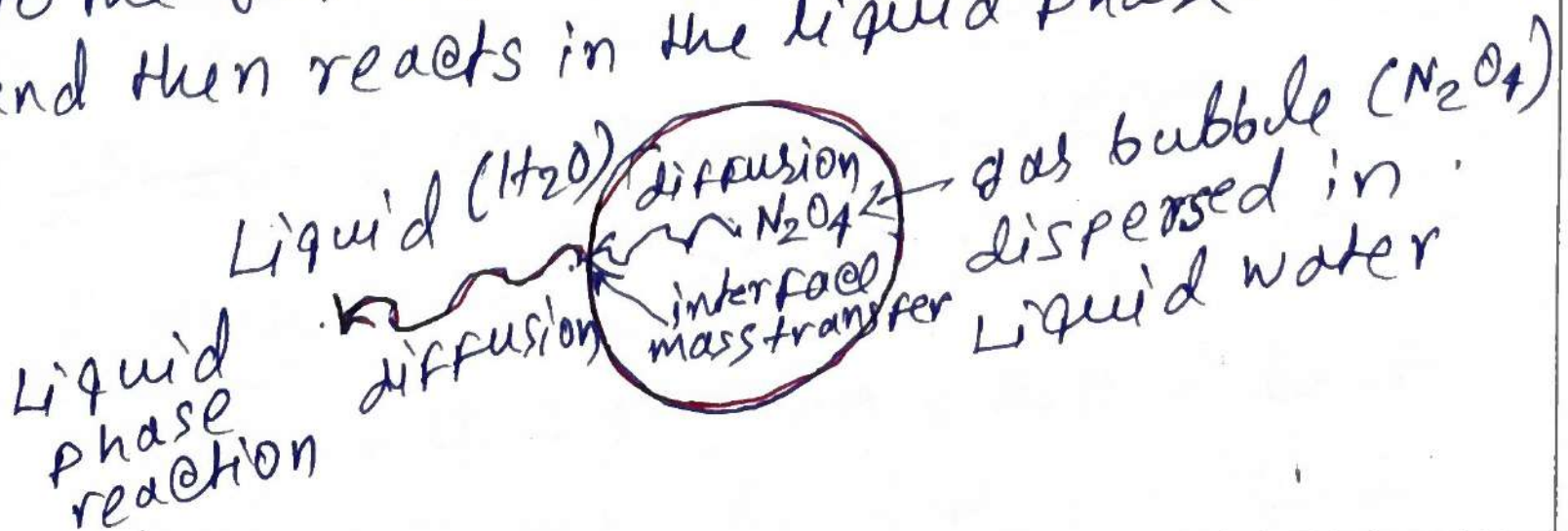
~~Catalytic~~ Non-catalytic heterogeneous reactions.
It is classified based on variety of phases involved.

- 1) Gas and Liquid.
- 2) Gas and Solid.
- 3) Liquid and Liquid.
- 4) Liquid and Solid.
- 5) Solid and Solid.

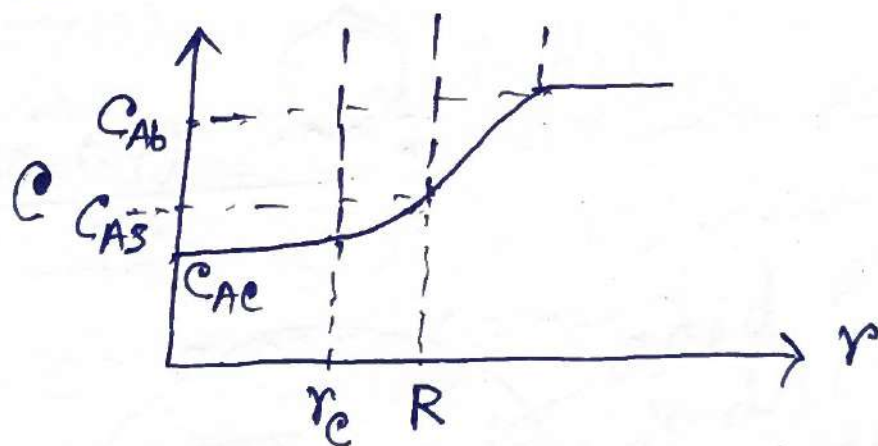
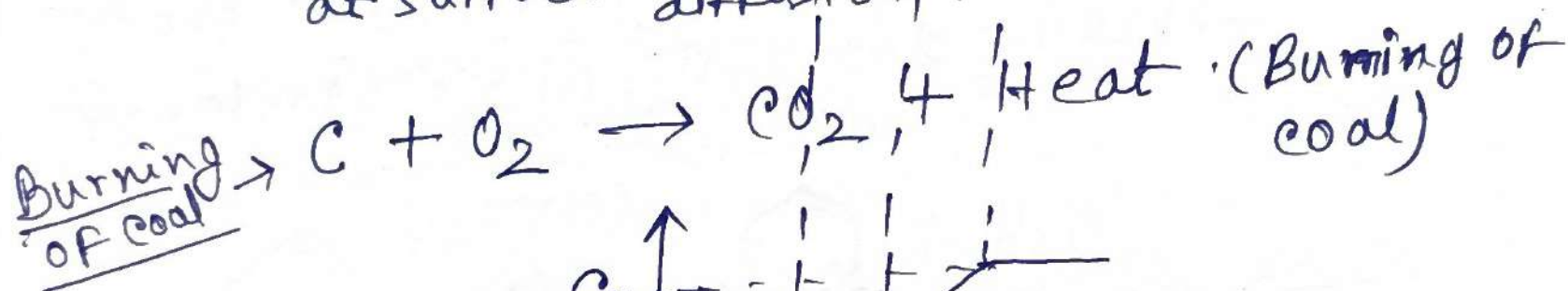
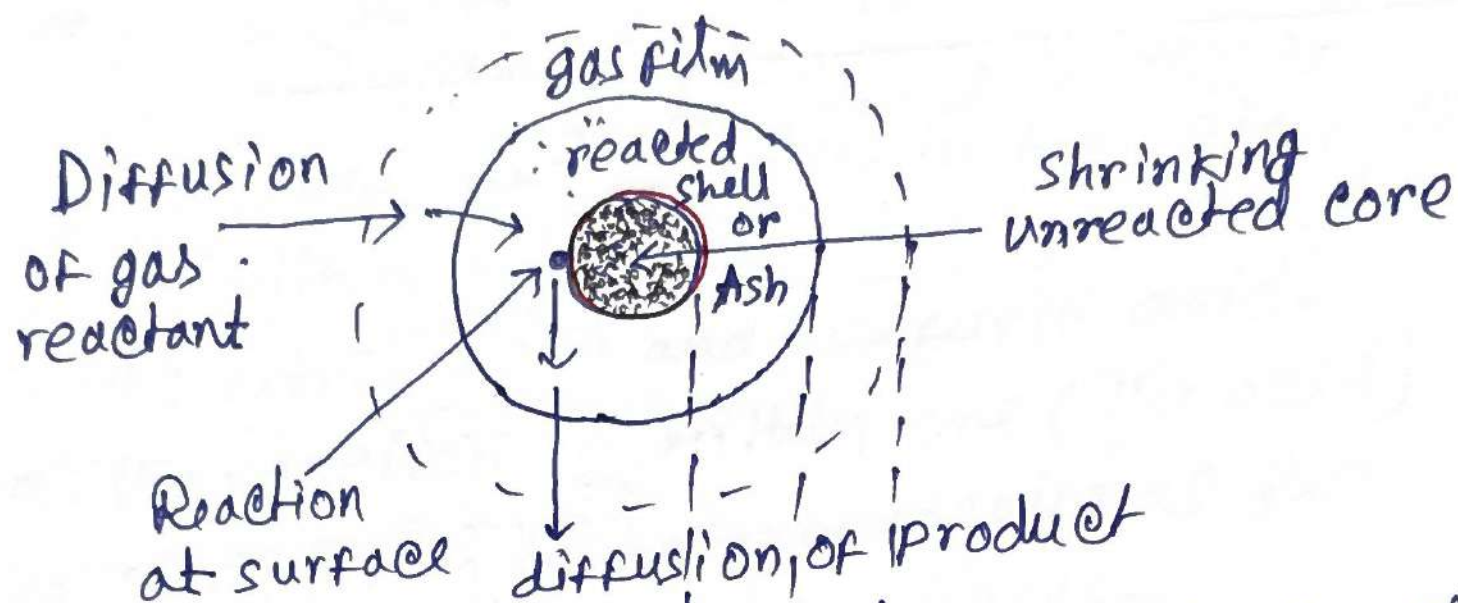
1) Gas-Liquid reaction: Reaction of nitrogen tetroxide (N_2O_4) with liquid water to produce nitric acid (HNO_3).



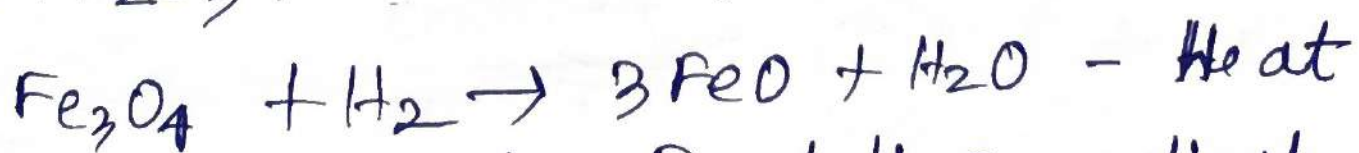
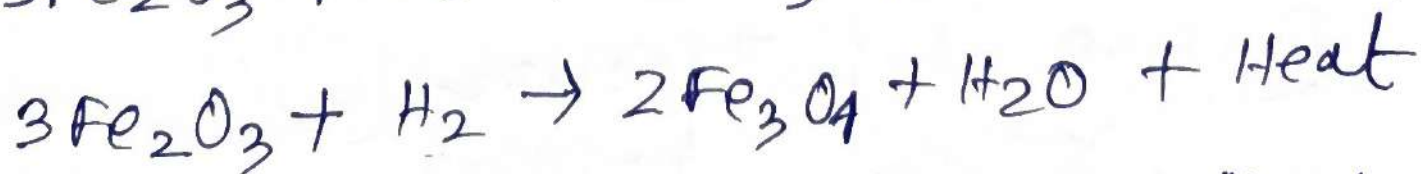
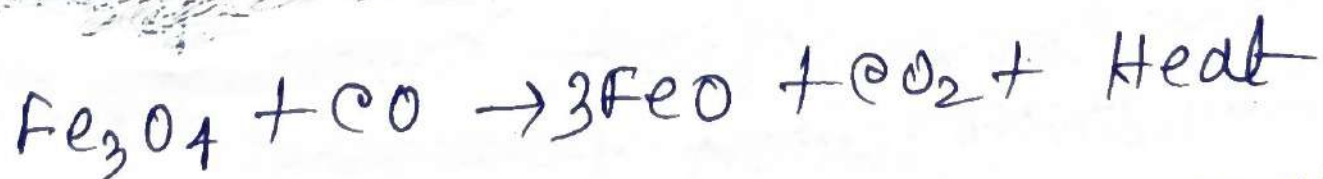
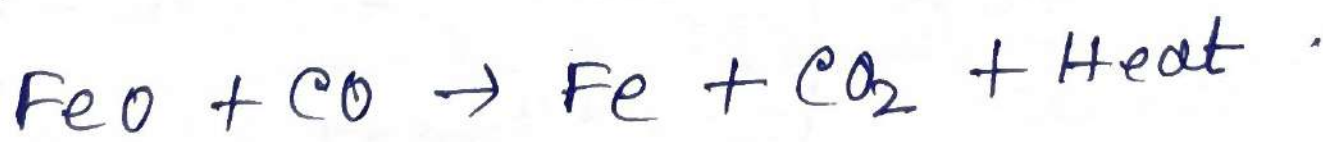
N_2O_4 (gas) is contacted with water (liquid). or dilute HNO_3 (aq.); the N_2O_4 diffuses to the gas-liquid interface, is absorbed, and then reacts in the liquid phase.



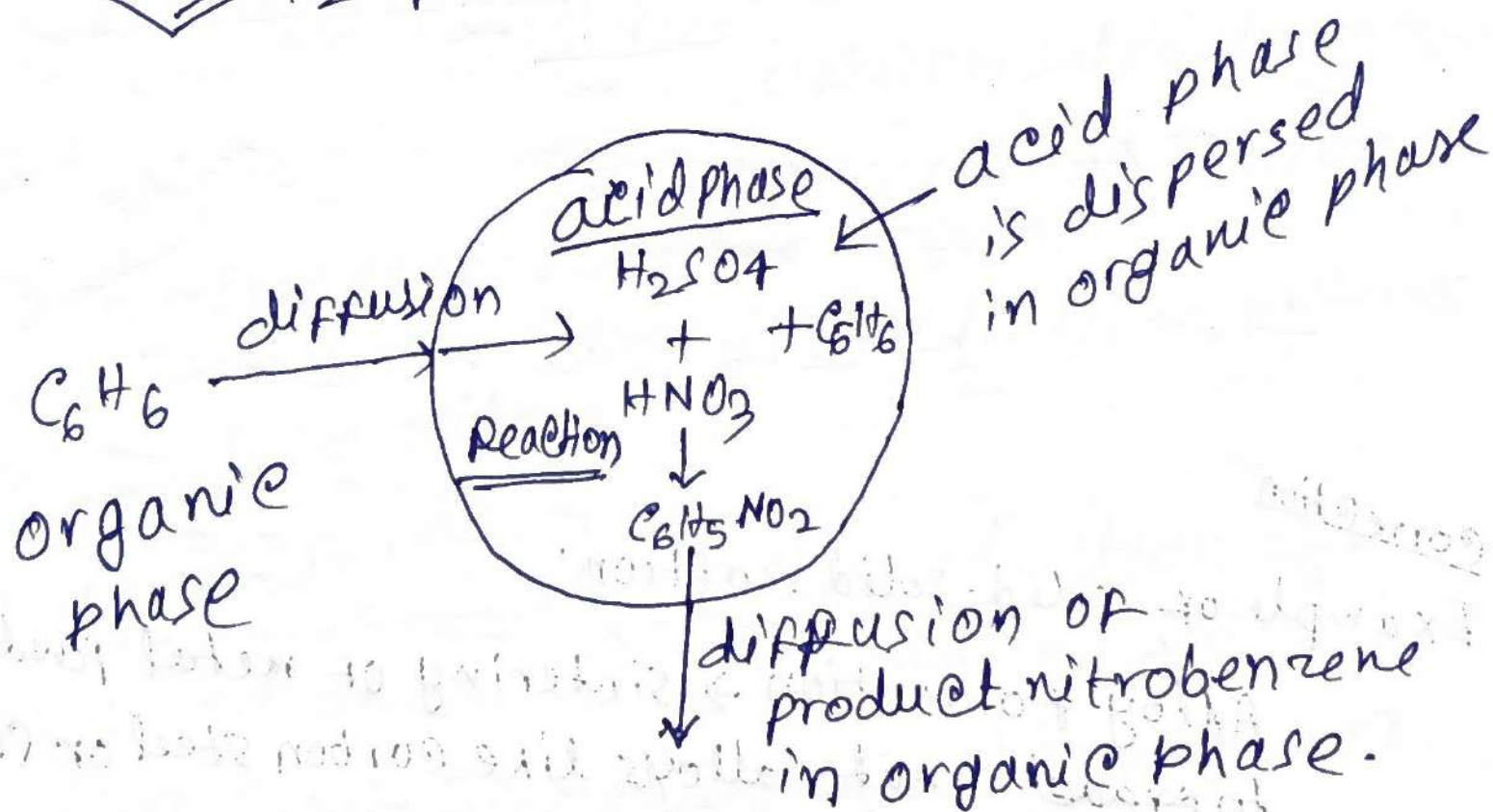
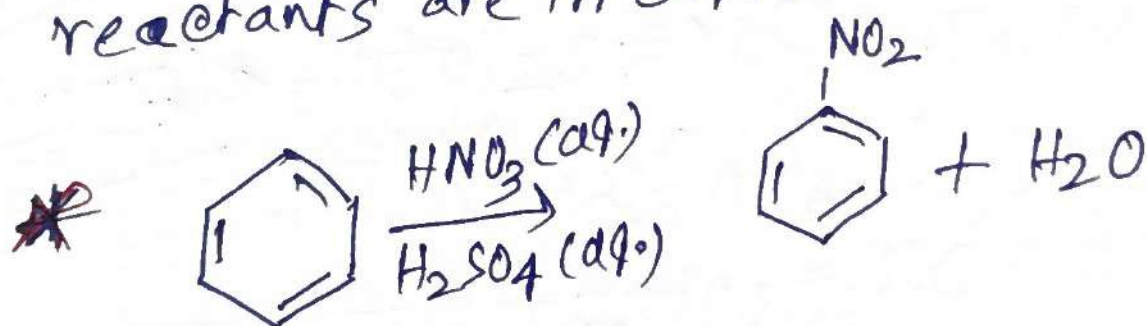
2) Gas-solid reaction: These types are burning of coal particles, ore reduction, the coking of catalyst particles, etc.



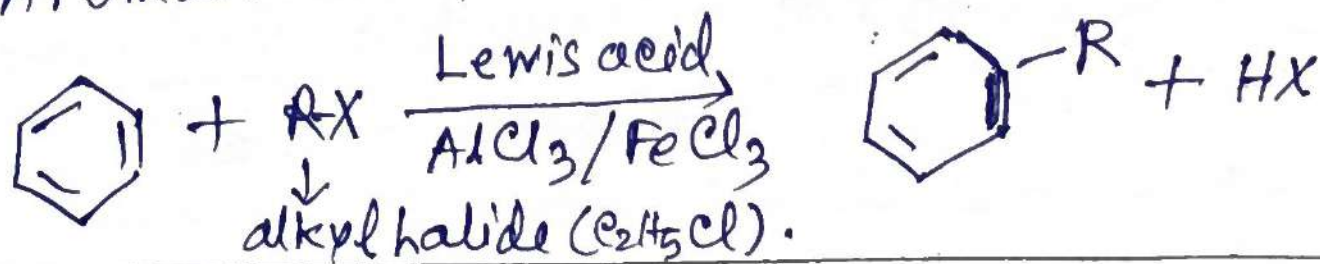
Iron Ore reduction



3) Liquid-liquid reaction: Nitration of benzene is liquid-liquid reaction. The benzene is intimately contacted with the aqueous phase containing a mixture of HNO_3 and H_2SO_4 . The acid (aq.) phase is immiscible with benzene. However, benzene diffuses and dissolves in the acid phase to react with a nitronium ion generated from the nitric acid and sulfuric acid. Here, the reaction is within one (the acid) phase. The system is heterogeneous as the reactants are in different phases.

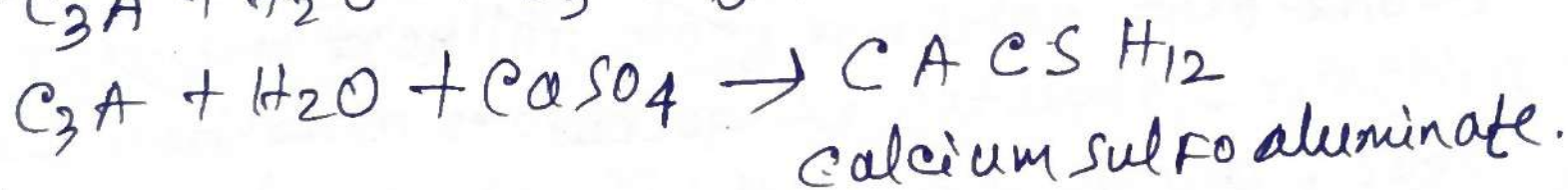
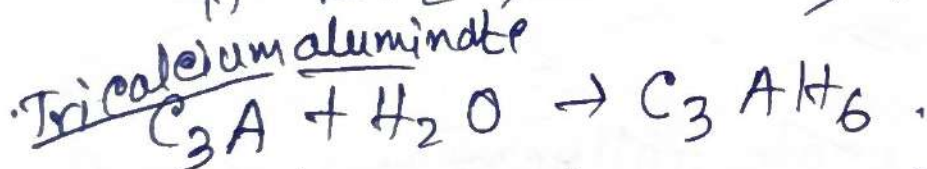
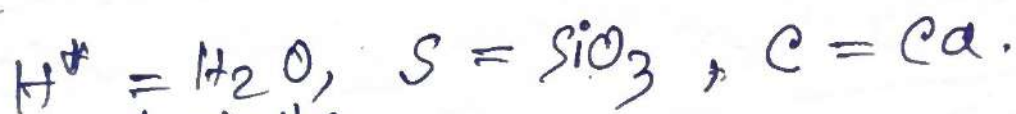
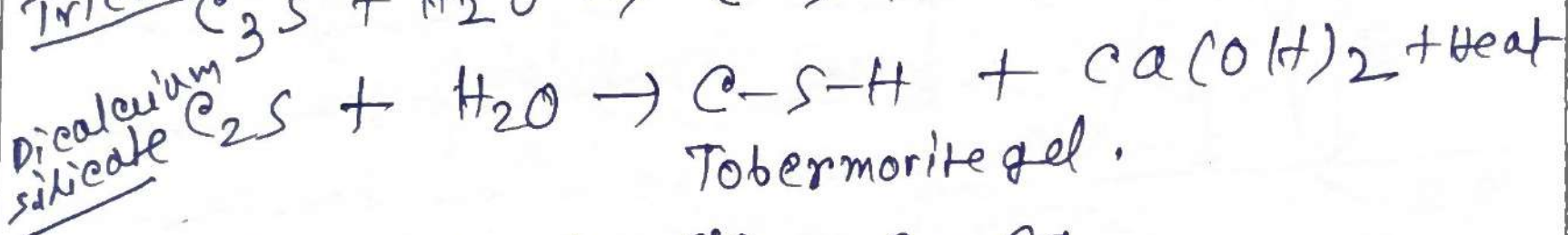
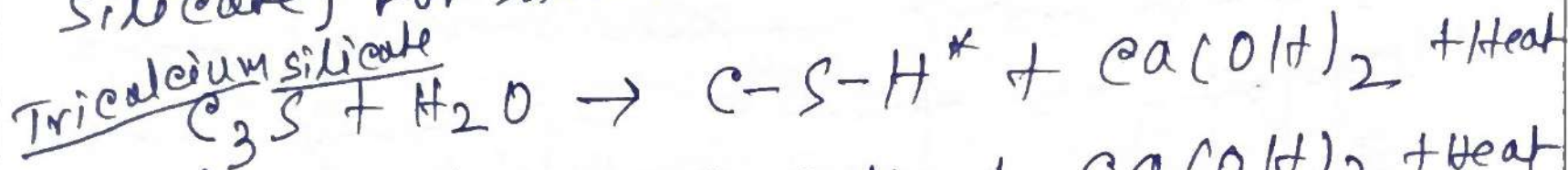


* Aromatic alkylation: [Friedel-Crafts]



4) Liquid-solid reaction: A liquid reacts with a solid substance at the interface. Moreover, the reactant can diffuse into the solid porous body and reacts with the solid substance at the porous surface.

Example: Hydration of cement (calcium silicate) for solidification.



Other

Example: Corrosion $\rightarrow$ rusting of iron ore in presence of water and oxygen.

Electrochemical reaction at electrodes in batteries or electrolytic cells.

5) Solid-solid reaction: The reduction of powdered oxides using a reducing solid phase.

Example: Mixture of metal oxide and carbon.

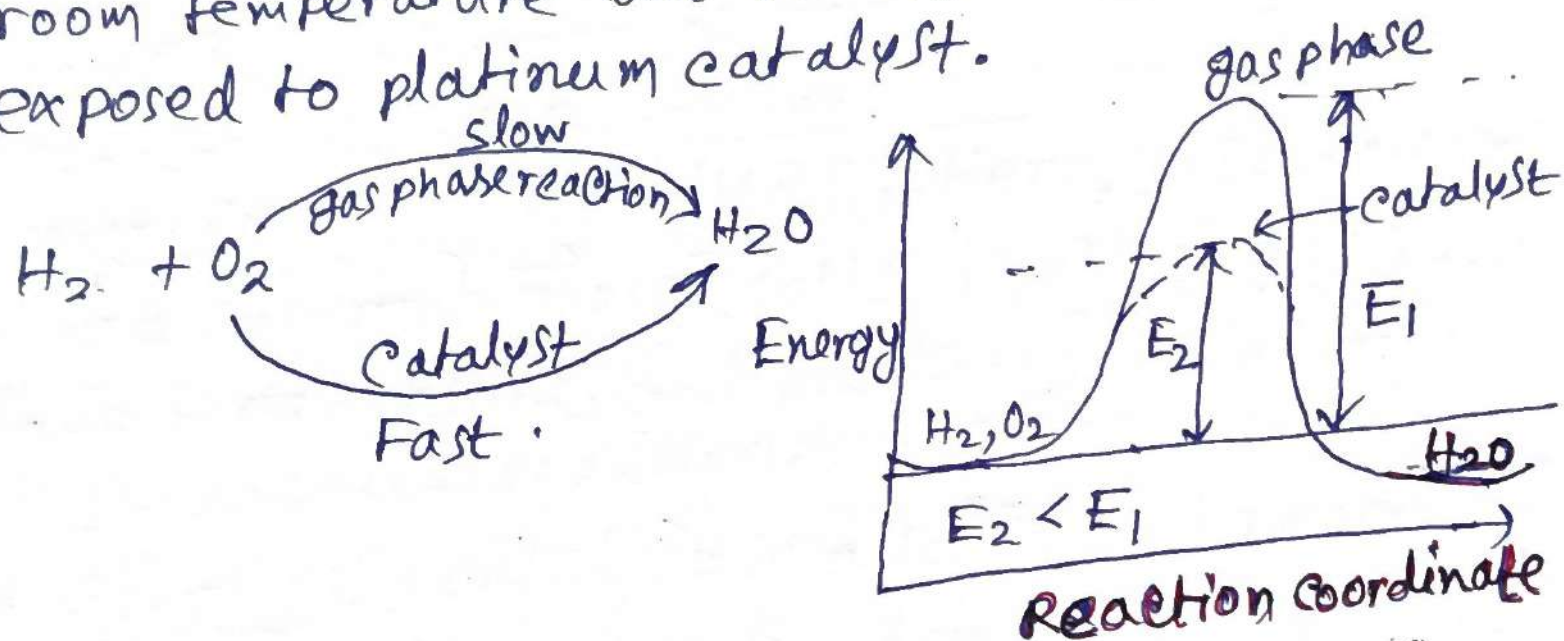
The iron-ore reduction by coke. The reaction between iron and sulfur to form iron sulfide.



Catalyst

Catalyst is a substance that affects the rate of a reaction but remains unchanged in the process.

Example: Gaseous H_2 and O_2 are inert at room temperature but react rapidly when exposed to platinum catalyst.



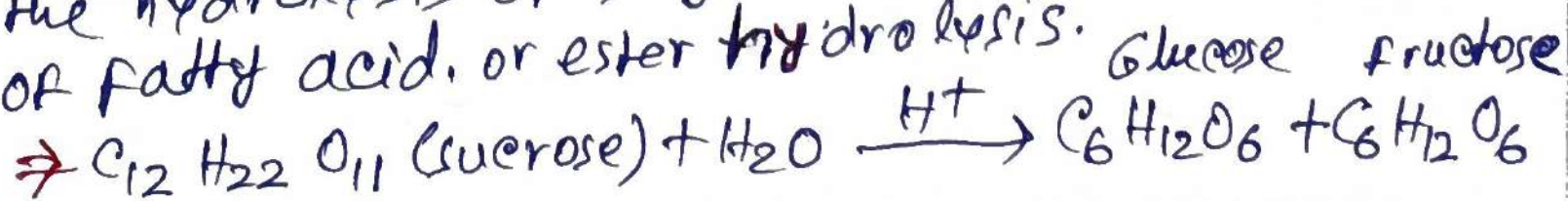
Progress of reaction along reaction path shows that activation energy of the catalytic reaction (E_2) is lower (much) than that of gas phase reaction without catalyst.

Homogeneous catalysis: Here, a catalyst is in solution with at least one of the reactants.

- * The catalyst and reactants are in the same phase (e.g. both in liquid or gaseous).

- * Homogeneous catalysts are often soluble in the reaction media.

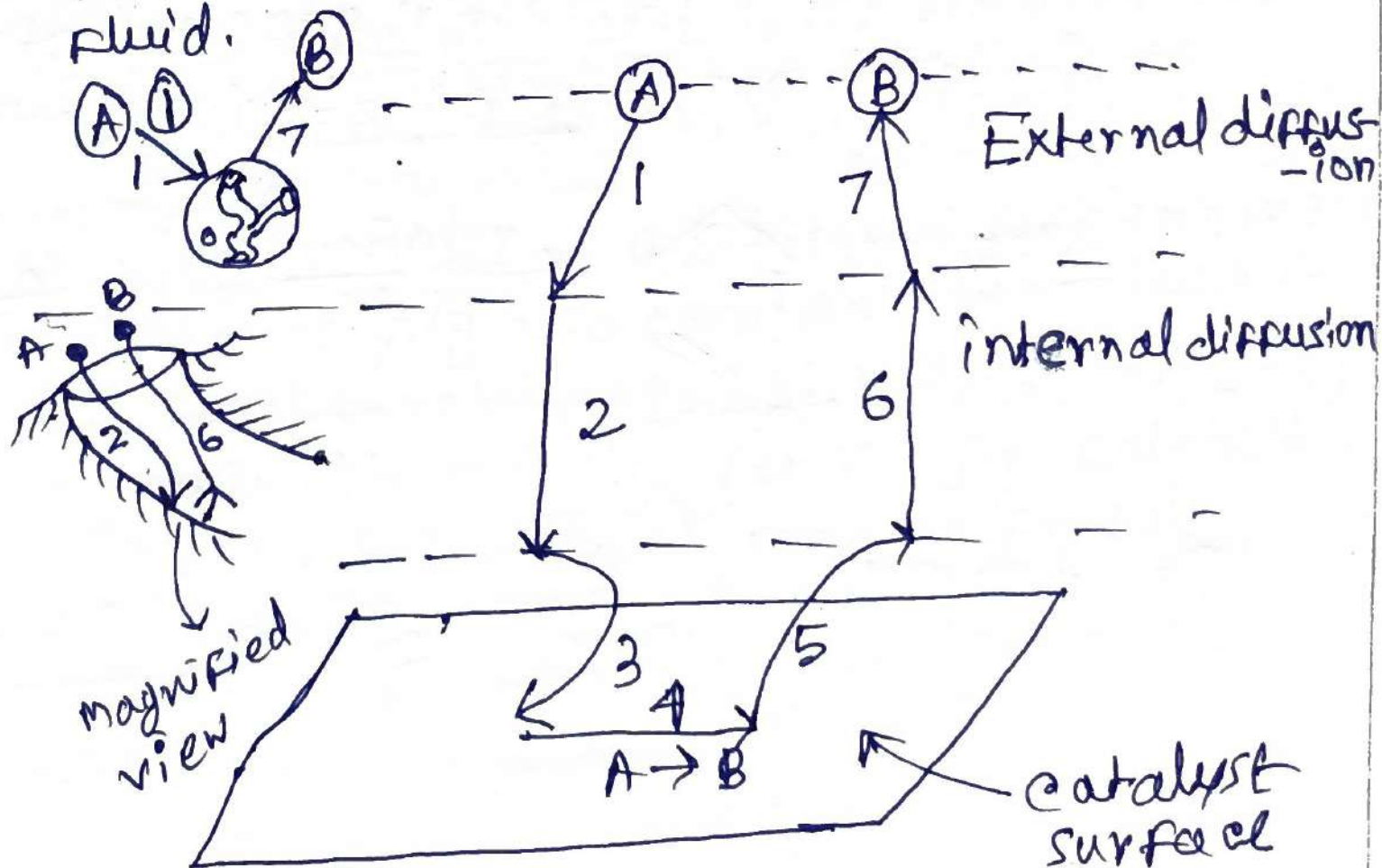
acid catalysis: sulfuric acid catalyzing the hydrolysis of sugar or esterification of fatty acid, or ester hydrolysis.



Gas phase homogeneous catalysis

Steps in catalytic reaction

1. Mass transfer (diffusion) of reactants (A) from the bulk fluid to the external surface of the catalyst pellet.
2. Diffusion of the reactant (A) from the pore mouth (catalyst pellet pore) to the internal catalyst surface.
3. Adsorption of reactant (A) onto the catalyst surface.
4. Reaction on the catalyst surface. (e.g. $A \rightarrow B$)
5. Desorption of the products (B) from the surface.
6. Diffusion of the products (B) from the interior of the pellet to the pore mouth (or external surface).
7. Mass transfer of the products (B) from the external pellet surface to the bulk fluid.



Catalyst pellet : In a packed bed reactor.

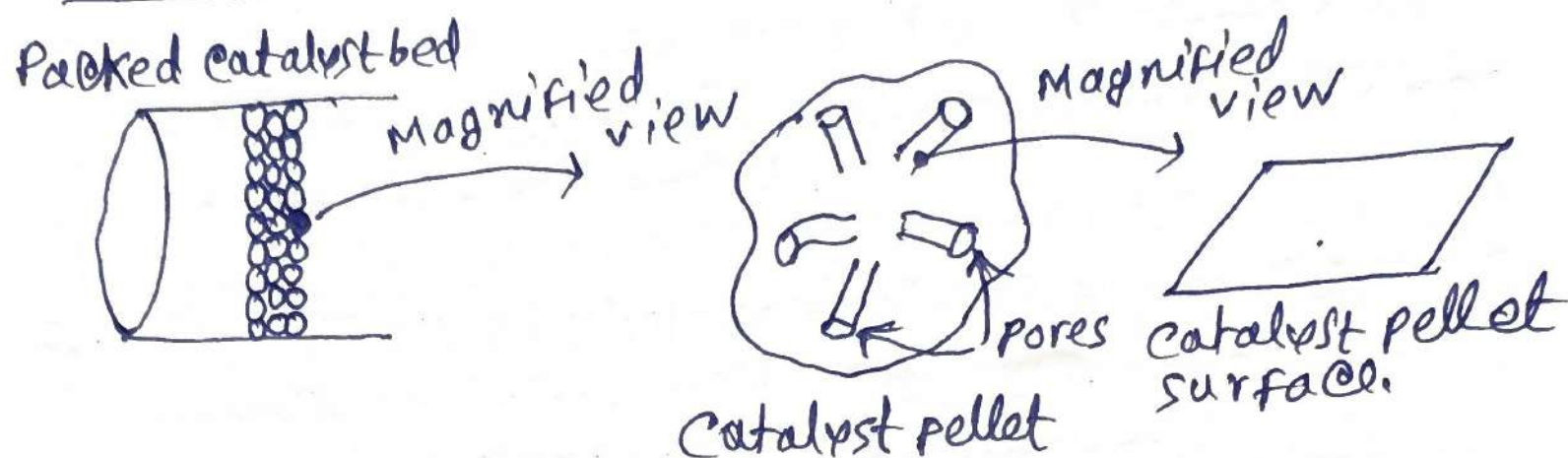


Fig: view of a catalyst pellet and its catalytic surface in a packed-bed reactor.

Components of the catalyst formulation

The catalyst pellet consists of the following :

- ① The support (catalyst support) is usually porous upon which a catalyst agent is deposited. Eg. Al_2O_3 , SiO_2 , or $\text{SiO}_2\text{-Al}_2\text{O}_3$
- ② The catalyst agent is the main catalyst material. Eg. Pt, V, Cu, Fe, etc.
- ③ The promoter is a substance that enhances the activity of a catalyst when added in small quantities. Eg. Molybdenum (Mo) acts as a promoter for the Fe-catalyst in the Haber-Bosch process for NH_3 synthesis.

- * Catalyst activity $\propto$ concentration of active site X.
- * The active sites are hosted on the pores of a support of high specific surface area.
- * The specific surface area ranges from 1 — several hundred square meter per gm.
- * α -alumina surface area (specific), $S_p = 10 \text{ m}^2/\text{gm}$
 η -alumina, $S_p = 100 - 300 \text{ m}^2/\text{gm}$
 γ -alumina, $S_p =$ do. (same as η -alumina)
- * Catalyst agents are deposited on the support.
 Eg. 0.5 wt% of Pt upon α - Al_2O_3 and γ - Al_2O_3 .
- * High surface area support is characterized by pores of small diameter.
- * Promoters are classified as physical or chemical.
 $\Rightarrow$ Physical promoter eg.: Addition of small quantities of Al_2O_3 to an Fe catalyst employed in NH_3 synthesis prevents the sintering of Fe crystallites.
 $\Rightarrow$ Chemical promoter eg.: Addition of K_2O to the same catalyst increases the intrinsic activity of the Fe crystallites.
 $\Rightarrow$ Negative chemical promoter or inhibitor eg.:

$$\text{C}_2\text{H}_4 + \frac{1}{2}\text{O}_2 \xrightarrow[\text{ethylene dichloride promoter (-ve)}]{\text{C}_2\text{H}_4\text{Cl}_2} \text{C}_2\text{H}_4\text{O} \text{ (ethylene oxide)}$$

$$\text{C}_2\text{H}_4 + 3\text{O}_2 \xrightarrow[\text{Without Promoter}]{\text{Without Promoter}} 2\text{CO}_2 + 2\text{H}_2\text{O}$$

Catalyst Preparation

* SO₂ oxydation catalyst (supported V₂O₅):
(active material)

⇒ At first, 8.1 parts of finely powdered V₂O₅ are dissolved in 5.15 parts of KOH in 300 parts of water.

⇒ Dilute HCl is added to neutralize the solution.

⇒ The solution is diluted with 1200 parts of water.

⇒ Then the following is added:

[140 parts potassium silicate solution
(21% SiO₂, 9.5% K₂O)

+
Support { 60 parts of kieselguhr and 10 parts of fibrous asbestos.

⇒ The mixture is warmed to 60°C–70°C.

⇒ Dilute HCl is added until the solution is weakly alkaline.

⇒ Zeolite and carrier mixture is pressed to remove excess liquid, dried, and compressed into granules.

⇒ Then heated in air at 300°C–400°C.

* Preparation of solution with promoter and active components → Impregnation of support material with this solution → Co-precipitation of active components and promoters onto the support
↳ Drying → Shaping (pellets, granules, or honeycomb)

Physical properties of catalyst.

1) Surface area of a catalyst material.

Ex: Raney nickel is highly porous and has large surface area, with areas as high as $1,500 \text{ m}^2/\text{g}$.

If it is held in the hand, the heat due to adsorption of O_2 can be felt immediately.

Example: Spray drying and other procedures

for manufacturing small particles can produce particles as small as 2 to 5 μm .

What size of particles would be necessary if the external surface is to be $100 \text{ m}^2/\text{g}$?

$\rho_p = 2.0 \text{ g/cm}^3$ (density).

$\therefore$ External surface area per unit volume of a spherical particle of diameter d_p is

$$\frac{\pi d_p^2}{\pi d_p^3/6} = \frac{6}{d_p}$$

$\therefore$ Surface area per gram of particles would be,

$$S_g = \frac{6}{\rho_p d_p} = \frac{6}{2 d_p} = \frac{6}{2 \times 2 \times 10^{-4}}$$

$$\text{Now! } d_p = 2 \mu\text{m} = 2 \times 10^{-4} \text{ cm}$$

$$S_g = 1.5 \times 10^4 \text{ cm}^2/\text{g} = 1.5 \text{ m}^2/\text{g}.$$

$$\Rightarrow d_p = \frac{6}{\rho_p S_g} = \frac{6}{2.0 \times (100 \times 10^4)} \text{ cm} = 0.03 \times 10^{-4} \text{ cm} = 0.03 \mu\text{m}$$

$\boxed{S_g = 100 \text{ m}^2/\text{g}}$

* Determination of surface area

The standard method for the measurement of catalyst area is based on the physical adsorption of a gas on the solid surface.

Ex: N₂ adsorbed at equilibrium at normal boiling point (-195.8°C) is measured over a range of N₂ pressures below 1 atm.

⇒ BET Method (Brunauer-Emmlett-Teller)

One molecular layer of N₂ adsorbed is identified in order to determine the area.

For example: a nickel-on-kieselguhr catalyst with surface area 205 m²/g as measured by N₂ adsorption.

Nickel atoms surface area is measured by chemisorbed H₂ (40 m²/g).

BET method is based on Langmuir isotherm

→ BET method is based on Langmuir isotherm.

Assumptions:

- 1) All the surface area of the catalyst has the same activity for adsorption or it is energetically uniform.
- 2) There is no interaction between adsorbed molecules. i.e., amount of adsorbed gas is independent of the rate of adsorption per site.
- 3) All the adsorption occurs by the same mechanism.
- 4) The extent of adsorption is less than one complete monomolecular layer of the surface.

Rate of adsorption; r_a (per unit of total surface area) $= kP(1-\theta)$

θ = fraction of the surface covered by the adsorbed molecule.

$1-\theta$ = fraction of bare surface.

p = partial pressure of gas molecule.

∴ Rate of desorption per unit of total surface area $r_d = k'\theta$

at equilibrium, $r_a = r_d$

$$\Rightarrow kP(1-\theta) = k'\theta$$

$$\Rightarrow kP(1-\theta) = \theta \quad \boxed{k/k' = K}$$

$$\theta = \frac{KP}{1+KP} = \frac{v}{v_m}$$

v = volume of gas adsorbed.

v_m = volume of gas adsorbed when all active sites are covered. (complete monomolecular layer).

$$\Rightarrow \frac{KP}{v} = \frac{1+KP}{v_m} = \frac{1}{v_m} + \frac{KP}{v_m}$$

$$\Rightarrow \boxed{\frac{P}{v} = \frac{1}{Kv_m} + \frac{P}{v_m}} \Rightarrow \boxed{\text{Monomolecular adsorption}}$$

For multilayer adsorption *

$$\rightarrow \boxed{\frac{P}{v(P_0 - P)} = \frac{1}{v_m c} + \frac{(c-1)P}{c v_m P_0}} *$$

$\therefore P_0$ = saturation or vapor pressure,

c = constant for the particular temperature and gas-solid system.

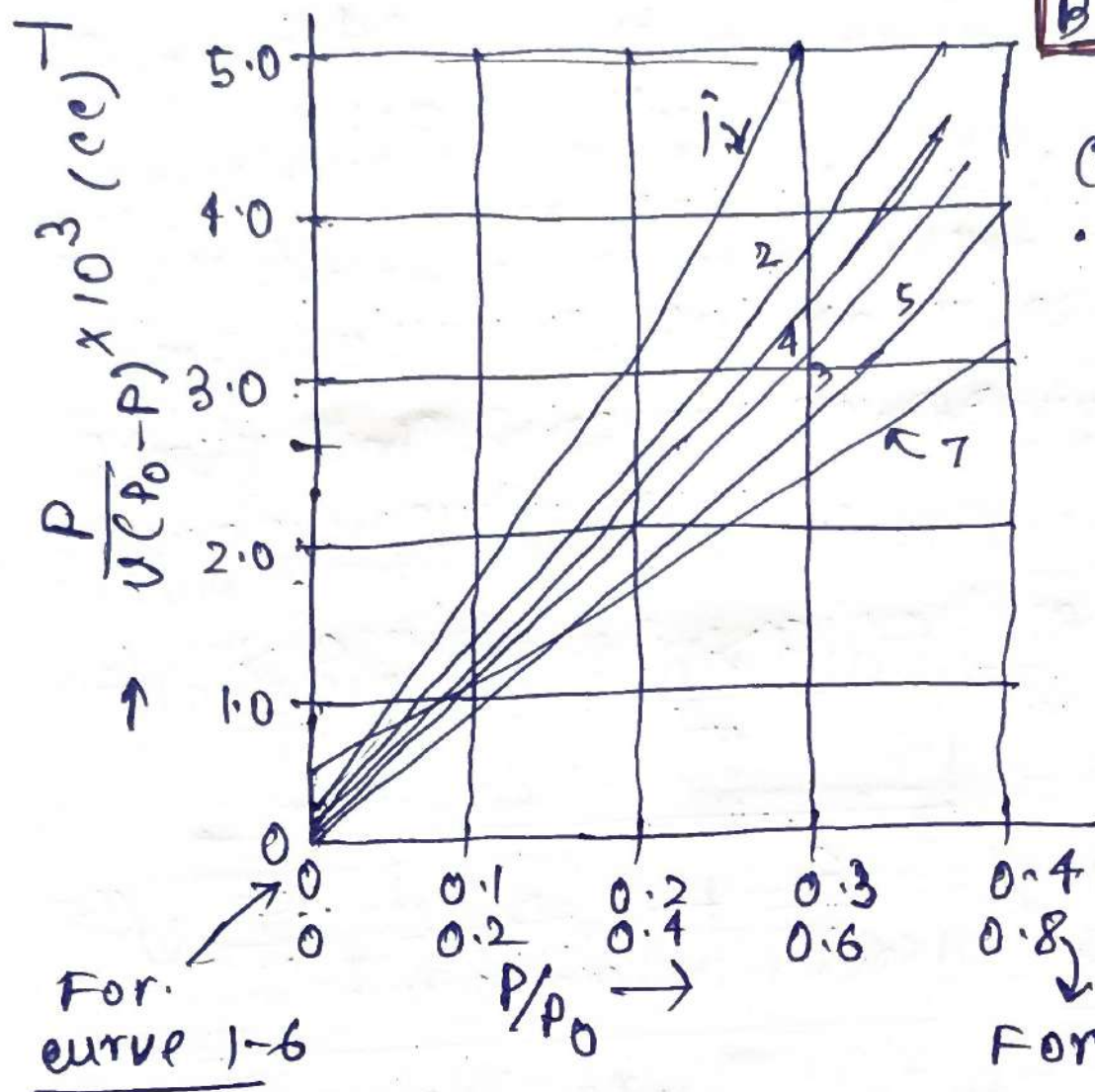
$\therefore \frac{P}{v(P_0 - P)}$ vs $\frac{P}{P_0}$ is a straight line.

intercept at $P/P_0 = 0$, $\boxed{I = \frac{1}{v_m c}}$

slope, $\boxed{S = \frac{c-1}{c v_m}}$

Solving these equation $\Rightarrow \boxed{v_m = \frac{1}{I+S}} *$

BET Plot



- Curve 1: CO_2 at -78°C
- 2: A at -183°C
- 3: N_2 at -183°C
- 4: O_2 at -183°C
- 5: CO at -183°C
- 6: N_2 at -195.8°C
- 7: nC_4H_{10} at 0°C .

(Data based upon 0.606g sample of silica)

* Total surface area per gram of solid adsorbent is, $S_g = \left[\frac{V_m N_0}{V} \right] \alpha$

N_0 = Avogadro's number, 6.02×10^{23} molecules/mole

V = volume per mole of gas at conditions of V_m .
= 22,400 $\text{cm}^3/\text{g mole}$. (molar volume)

V_m is based on a 1.0 gm sample.

α = Projected area of a molecule on the surface (two-dimensional packing of molecules on the surface).

$$\alpha = 1.09 \left[\frac{M}{N_0 \rho} \right]^{2/3}; \quad M = \text{molecular wt.}$$

ρ = density.

Ex 1: For N_2 at $-195.8^\circ C$ (boiling point)
 - $\rho = 0.808 \text{ g/cm}^3$ [0.606 g of catalyst]

$$\alpha = 16.2 \times 10^{-16} \text{ cm}^2 = 16.2 \text{ \AA}^2$$

$$S_g = 4.35 \times 10^4 \text{ cm}^2/\text{g solid catalyst}$$

From BET Plot (curve 6) N_2 at $-195.8^\circ C$

$$\text{Intercept, } I = 0.1 \times 10^{-3} \text{ cm}^3$$

$$S = \frac{5.3 - 0.1}{0.4 - 0} = 13 \times 10^{-3} \text{ cm}^3$$

$$U_m = \frac{1}{I + S} = \frac{1}{(0.1 + 13) \times 10^{-3}} \frac{\text{cm}^3}{(0.606) \text{ g catalyst}}$$

$$= 126 \text{ cm}^3/\text{g catalyst}$$

$$S_g = 4.35 \times 10^4 \times 126 \text{ cm}^2/\text{g catalyst}$$

$$S_g = 4.35 \times 126 \text{ m}^2/\text{g catalyst} = 550 \text{ m}^2/\text{g catalyst}$$

Ex 2: Estimate the surface area of the silica gel by using the adsorption data for O_2 at $-183^\circ C$. $\rho_{O_2}(\text{liquefied}) = 1.14 \text{ g/cm}^3$ at $-183^\circ C$.

$$\alpha = 1.09 \left[\frac{32}{6.02 \times 10^{23} \times 1.14} \right]^{2/3} = 14.2 \times 10^{-16} \text{ cm}^2$$

$$S_g = \frac{U_m (6.02 \times 10^{23})}{22,400} 14.2 \times 10^{-16} = 3.8 \times 10^4 \text{ cm}^2/\text{g}$$

From BET Plot (curve 4)

$$I = 0.4 \times 10^{-3} \text{ cm}^3, S = \frac{5.4 - 0.4}{0.38 - 0} \times 10^{-3} = 13.2 \times 10^{-3} \text{ cm}^3$$

$$U_m = \frac{10^3}{0.4 + 13.2} \frac{1}{0.606} = 122 \text{ cm}^3/\text{g of catalyst}$$

$$S_g = 3.8 \times 10^4 (122) = 465 \times 10^4 \text{ cm}^2/\text{g}$$

$$S_g = \underline{\underline{465 \text{ m}^2/\text{g}}}$$

* Determination void volume (porosity) and solid density:

void volume or porosity of a catalyst particle can be estimated by,

⇒ weight a catalyst sample → immersed in a liquid water

The sample is superficially dried and weighed.

↓
Boiling the sample to displace the air in the pores.

* The increase in weight = pore volume.
Liquid density (water)

Helium-Mercury method: The volume of helium displaced by a sample of catalyst is measured. The helium is then removed by the volume of mercury is measured. As the mercury not fills the pores at atmospheric pressure, the difference in volume gives the pore volume.

$$\text{porosity, } \epsilon_p = \frac{\text{Void volume}}{\text{Total volume}} = \frac{m_p V_g}{m_p V_g + m_p / \rho_s}$$

$$\Rightarrow \boxed{\epsilon_p = \frac{V_g \rho_s}{V_g \rho_s + 1}}$$

m_p = mass of the particle

V_g = void volume per gram of particles.

ρ_s = density of the particle.

In mercury method:

$$\epsilon_p = \frac{V_g}{1/\rho_p} = \rho_p V_g$$

$\Rightarrow \rho_p$ = density of the porous particle (void + solid)
= $\frac{\text{Mass of the particles sample}}{\text{Mercury volume}}$

$\Rightarrow \rho_p$ = Mass of the particle per volume of sample.

Example 8-4 In an experiment to determine the pore volume and porosity of a catalyst-particle. The following data were obtained on a sample of activated silica (granular, 4 to 12 mesh size):

Mass of catalyst sample = 101.5 g

Volume of helium displaced by sample = 45.1 cm³

Volume of Mercury " " " = 82.7 cm³

calculate required property:

Ans: $V_g = \frac{82.7 - 45.1}{101.5} = 0.371 \text{ cm}^3/\text{gm}$

$\rho_s = \frac{m_p}{\text{volume of helium displaced by sample}}$
density of solid particle.

$$\rho_s = \frac{101.5}{45.1} = 2.25 \text{ g/cm}^3$$

$$\epsilon_p = \frac{V_g \rho_s}{V_g \rho_s + 1} = \frac{0.371 \times 2.25}{0.371 \times 2.25 + 1} = \boxed{0.455}$$

Example 8.5 A hydrogenation catalyst is prepared by soaking alumina particles (100 to 150 mesh) in aqueous NiNO_3 solution. After drying and reduction, it contains about 7 wt% NiO . It is then made into large cylindrical pellets. The gross measurements for one pellet are:

Mass = 3.15 g.

Mass = 3.15 g.
Diameter = 1.00 in; Thickness = $\frac{1}{4}$ in;

Diameter = 1.00 cm
 volume = 3.22 cm³

Diameter = 1.00 mm
 Volume = 3.22 cm³
 The micropore volume of the pellet = 0.645 cm³
 The micropore " " " " = 0.40 cm³/gm (2d)

The micropore of particles. (void + solid)

a) Density of the pellet, $\rho_p = \frac{3.15}{3.22} = 0.978 \text{ g/cm}^3$ (void + solid).

b) Macropore volume per gram:

$$(v_g)_m = \frac{0.645}{3.15} = 0.205 \text{ cm}^3/\text{g}$$

c) Macropore void fraction, ϵ_m

$$\epsilon_M = (V_g)_M S_F = 0.205 \times 0.978$$

$$\epsilon_M = 0.20$$

d) Micropore void fraction ϵ_u

$$e_u = (v_g)_u s_p = 0.4 \times 0.978 = \boxed{0.391}$$

e) Solid fraction = $1 - e_m - e_x = 1 - 0.2 - 0.391$

$$e_s = 0.409 \quad \text{as} \quad e_m + e_u + e_s = 1$$

f) Density of the particle (For micropore + solid volume).

$$\rho_{p,u} = \frac{3.15}{3.22 - 0.645} \quad \frac{\text{gm of particle per}}{\text{gm of microporous solid.}}$$

$$\rho_{p,u} = 1.22 \text{ g/cm}^3$$

g) Density of the solid phase.

$$\rho_s = \frac{\text{Mass of pellet}}{(\text{Volume of pellet}) \times \epsilon_s}$$

$$= \frac{V_p \rho_p}{V_p \epsilon_s} = \frac{\rho_p}{\epsilon_s} = \frac{0.978}{0.409} = \underline{\underline{2.39 \text{ g/cm}^3}}$$

h) void fraction of particles.

$$\epsilon_p = \frac{(V_g)u}{V_{sp}} = (V_g)u \rho_{p,u} = 0.39 \times 1.22 = \underline{\underline{0.49}}$$

$$\underline{\text{but total void}} \Rightarrow \epsilon_m + \epsilon_u = \underline{\underline{0.591}}$$

$$\text{and } \epsilon_s = 0.409 \text{ (solid).}$$

* Determination of pore volume distribution

Effectiveness of internal surface for catalytic reaction = function (ϵ_p, V_g , pore-volume distribution)

Pore-volume distribution or distribution of void volume in catalyst mass is based on the sizes of pore openings.

* In electron-microscope (scanning) photographs
the darker regions represent void space between individual particles, and smaller dark regions are intraparticle void. The light portions are solid silver for the porous silver particles ($S_g = 19.7 \text{ m}^2/\text{g}$).

* Mercury-penetration method.

The force due to surface tension tends to keep mercury out of a pore.



Mercury penetration in a pore of radius a .

$$a = \frac{-2\gamma \cos \theta}{p}$$

* Darke (1945)

γ = Surface tension of mercury.

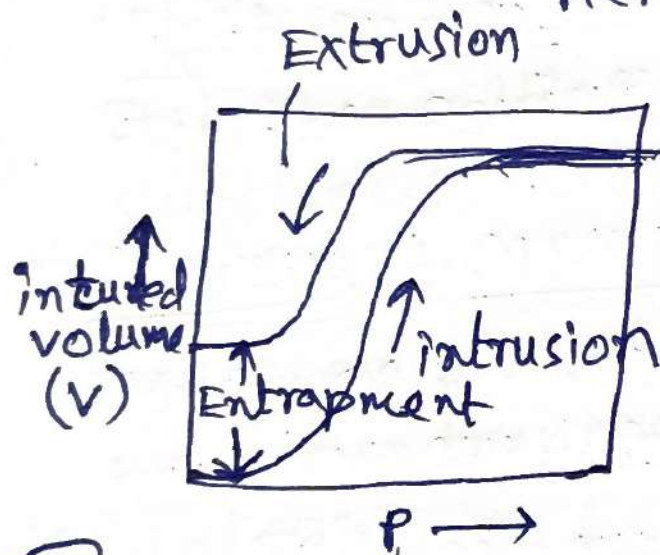
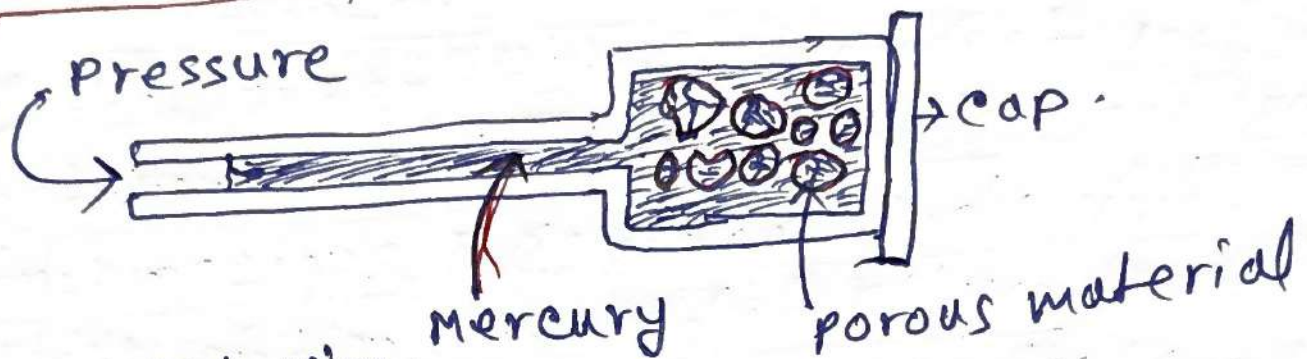
θ = contact angle between the mercury and pore wall.

p = Pressure (applied)

Methodology of Mercury intrusion porosimetry (MIP)

It is used to characterize the pore structure of materials by forcing mercury into the pores under increasing pressure.

The amount of mercury that enters the pores at different pressure level is measured to determine the pore size distribution, total pore volume, and other porosity parameters.



Pore volume distribution
in VO_2 pellet.
distribution curve

